

# M51 Whirlpool and NGC 1316 Fornax A -- MUGE with Simulation Scripts

Daniel T. Murphy

## PAPER\_750: M51 Whirlpool and NGC 1316 Fornax A — MUGE with Simulation Scripts

**Author:** Daniel T. Murphy

**Framework:** Universal Quantum Field Superconductive Framework (UQFF)

**Session:** 180 continuation | v5.38

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**CP4 Class:** #334 — M51NGC1316MUGESimulationCalculator

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### Abstract

Two Hubble-studied galaxies — M51 (the Whirlpool Galaxy, interacting with NGC 5195) and NGC 1316 (Fornax A, a post-merger elliptical) — represent opposite ends of the galaxy evolution spectrum.

This paper derives tailored Master Universal Gravity Equations (MUGEs) for both systems using Hubble ACS datasets, incorporating interaction-specific environmental terms:  $F_{\text{tidal}}$  and  $\psi_{\text{spiral}}$  for M51, and  $F_{\text{tidal}} + F_{\text{cluster}} + \rho_{\text{dust}}$  for NGC 1316. Full Python simulation scripts (m51\_simulation.py, ngc1316\_simulation.py) are included for radial acceleration profile generation.

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### 1. Introduction

**1.1 M51 (Whirlpool Galaxy)** M51 is a grand design spiral at 7.7 Mpc with an active tidal interaction companion NGC 5195. Hubble ACS observations (2005) provide: -  $M_{\text{visible}} = 1.2 \times 10^{11} \text{ MM}_{\text{sun}}$  -  $M_{\text{DM}} = 4 \times 10^{10} \text{ MM}_{\text{sun}}$  -  $\text{SFR} \approx 1 \text{ MM}_{\text{sun}}/\text{yr}$  -  $M_{\text{BH}} \approx 10^6 \text{ MM}_{\text{sun}}$  - Distance = 7.7 Mpc,  $z \approx 0.0015$  -  $B \approx 5 \times 10^{-6} \text{ T}$

**1.2 NGC 1316 (Fornax A)** NGC 1316 is a giant elliptical at 75 Mpc with a complex merger history: -  $M_{\text{total}} \approx 5 \times 10^{11} \text{ MM}_{\text{sun}}$  (visible + DM) -  $M_{\text{DM}} = 1.5 \times 10^{11} \text{ MM}_{\text{sun}}$  -  $M_{\text{BH}} \approx 10^8 \text{ MM}_{\text{sun}}$  - Merger age: 1-3 Gyr ago (from ripples, dust lanes) -  $\rho_{\text{dust}} \approx 10^{-2} \text{ kg/m}^3$

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### 2. M51 Whirlpool Galaxy MUGE

\$\$

$\begin{aligned}$

$\& g_{\text{M51}}(r, t) = (GM(t)) / (r(t)^2) (1+H(t, z)) (1-B(t)/B_{\text{crit}}) (1+F_{\text{env}}(t)) \backslash \backslash$

$\& + (U_{\text{g1}} + U_{\text{g2}} + U_{\text{g3}} + U_{\text{g4}}) + U_{\text{i}} \backslash \backslash$

$\& + (\text{Lambdac}^{2/3}) \backslash \backslash$

```

& + (\hbar/\sqrt{\Delta t \Delta t_p}) \int (\psi_{total} / H \psi_{total} dV) (2\pi / t_{Hubble}) \\
& + \rho_{fluid} V g \\
& + (M_{vis} + M_{DM}) (\Delta \rho / \rho + (3G^* M) / r^3) \\
\end{aligned}

```

> **Canonical note:** The  $(G^* M(t)) / (r(t)^2)$  term is the Step 10 Newton observational projection — per UQFF chain, the foundational gravity seed is DPM/Ug1 ( $k_1 \mu M/r$ ). The Ug\_i terms in the equation are the canonical DPM-derived components.

### M51 F\_env Terms

```


```

### M51 H(t,z)

```


```

### M51 Parameters Summary

| Parameter | Value                                                  |
|-----------|--------------------------------------------------------|
| M_visible | 1.2x10 <sup>11</sup> MM_sun = 2.39x10 <sup>41</sup> kg |
| M_DM      | 4x10 <sup>10</sup> MM_sun = 7.96x10 <sup>40</sup> kg   |
| r_galaxy  | 30 kpc = 9.26x10 <sup>20</sup> m                       |
| B         | 5x10 <sup>-6</sup> T                                   |
| SFR       | 1 MM_sun/yr = 6.3x10 <sup>22</sup> kg/yr               |
| F_tidal   | 5.59x10 <sup>-13</sup> m/s <sup>2</sup>                |

## 3. M51 Simulation Script (m51\_simulation.py)

```

python
import numpy as np
import matplotlib.pyplot as plt

```

**Constants**  $G = 6.6743\text{e-}11 \text{ # m}^3 \text{ kg}^{-1} \text{ s}^{-2}$   $H_0 = 70\text{e}3 / 3.086\text{e}22 \text{ # s}^{-1}$   $\Lambda = 1.1\text{e-}52 \text{ # m}^{-2} \text{ c} = 3\text{e}8 \text{ # m/s}$   $\hbar = 1.0546\text{e-}34 \text{ # J.s}$   $t_{\text{Hubble}} = 4.35\text{e}17 \text{ # s}$   $B_{\text{crit}} = 1\text{e}11 \text{ # T}$   $\rho_{\text{vac\_SCm}} = 7.09\text{e-}37 \text{ # J/m}^3$   $\rho_{\text{vac\_UA}} = 7.09\text{e-}36 \text{ # J/m}^3$   $\mu_0 = 4 * \text{np.pi} * 1\text{e-}7 \text{ # H/m}$   $M_{\text{sun}} = 1.989\text{e}30 \text{ # kg}$   $\text{kpc} = 3.086\text{e}19 \text{ # m}$

**M51 Parameters**  $M_{\text{vis}} = 1.2\text{e}11 * M_{\text{sun}}$   $M_{\text{DM}} = 4.0\text{e}10 * M_{\text{sun}}$   $M_{\text{NGC5195}} = 1\text{e}10 * M_{\text{sun}}$   $r_0 = 30\text{e}3 * \text{kpc}$   $d_{\text{inter}} = 50\text{e}3 * \text{kpc}$   $B = 5\text{e-}6 \text{ # T}$   $\sigma_{\text{spiral}} = 1\text{e}3 * \text{kpc}$   $z = 0.0015$   $\lambda_I = 1.0$   $F_{\text{RZ}} = 0.01$   $\omega_i = 1.585\text{e-}8 \text{ # rad/s}$

```
def M_total():
    return M_vis + M_DM
```

```
def H_z(z):
    return H_0 np.sqrt(0.3 (1 + z)**3 + 0.7)
```

```
def F_tidal():
    return G M_NGC5195 / d_inter**2
```

```
def U_i():
    return lambda_I (rho_vac_SCM / rho_vac_UA) omega_i 1.0 (1 + F_RZ)
```

```
def psi_spiral(r):
    A = 1e-10
    return A np.exp(-r2 / (2 sigma_spiral**2))
```

```
def g_M51(r, t=0):
    M = M_total()
    H = H_z(z)
    F_env = F_tidal() / (G M / r2) # normalized
    g_grav = (G M / r**2) (1 + H) (1 - B / B_crit) (1 + F_env)
    quantum = hbar / np.sqrt(1e-10 1e-20) psi_spiral(r)**2 (2 np.pi / t_Hubble)
    dm_term = (M_vis + M_DM) (1e-5 + 3 G M / r**3)
    cosmological = Lambda c**2 / 3
    return g_grav + U_i() + cosmological + quantum + dm_term
```

**Compute radial profile**  $r_{\text{vals}} = \text{np.linspace}(1\text{e}3 * \text{kpc}, 30\text{e}3 * \text{kpc}, 200)$   $g_{\text{vals}} = [g_{\text{M51}}(r) \text{ for } r \text{ in } r_{\text{vals}}]$

```
plt.figure(figsize=(10, 6))
plt.plot(r_vals / kpc, g_vals, 'b-', linewidth=2, label='M51 MUGE Acceleration')
plt.xlabel('Radius (kpc)')
plt.ylabel('g_M51 (m/s^2)')
```

```

plt.title('M51 Whirlpool Galaxy — UQFF MUGE Radial Profile')
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig('m51_gravity_profile.png', dpi=150)
plt.close()
print("M51 simulation complete. Profile saved to m51_gravity_profile.png")
'''

```

---

## 4. NGC 1316 (Fornax A) MUGE

```

''
g_NGC1316(r, t) = (GM(t)) / (r(t)^2) (1+H(t,z)) (1-B(t)/B_crit) (1+F_env(t))
+ (U_g1 + U_g2 + U_g3' + U_g4) + U_i
+ (Lambdac^2/3)
+ (hbar/√(DeltaxDeltap)) integral(psi_totalHpsi_total dV) (2pi/t_Hubble)
+ rho_dustVg
+ (M_vis + M_DM) (deltarho/rho + (3G*M)/r^3)
''

```

### NGC 1316 $F_{env}$ Terms

'''

$F_{env}(t)_{NGC1316} = F_{tidal} + F_{cluster}$

$F_{tidal}$ : tidal from past spiral galaxy mergers

$F_{tidal} = G \cdot M_{spiral} / d_{spiral}^2$

$M_{spiral} \sim 10^{10} M_{sun}$  (remnant progenitor)

$d_{spiral} \sim 50$  kpc

$F_{tidal} \sim 5.59 \times 10^{-13} \text{ m/s}^2$

$F_{cluster}$ : star cluster tidal disruption

$F_{cluster} = k_{cluster} \cdot M_{cluster}$

$k_{cluster} \sim 10^{-12} \text{ N/MM}_{sun}$

$M_{cluster} \sim 10^6 M_{sun}$

$F_{cluster} \sim 10^{-6} \text{ m/s}^2$

$\rho_{dust}$ : dust lane fluid term

$\rho_{dust} = 10^{-21} \text{ kg/m}^3$  (ACS-derived dust column density)

'''

### NGC 1316 Mass Evolution

''

$M(t) = M_{vis} + M_{DM} + M_{merger}(t)$

$M_{merger}(t) = M_{spiral} \cdot e^{(-t/\tau_{merge})}$

$\tau_{merge} \sim 1 \text{ Gyr} = 3.156 \times 10^{16} \text{ s}$

''

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## 5. NGC 1316 Simulation Script (ngc1316\_simulation.py)

```
```python
import numpy as np
import matplotlib.pyplot as plt
```

**Constants (same as M51)  $G = 6.6743\text{e-}11$   $H_0 = 70\text{e}3 / 3.086\text{e}22$   $\Lambda = 1.1\text{e-}52$   $c = 3\text{e}8$   $\hbar = 1.0546\text{e-}34$   $t_{\text{Hubble}} = 4.35\text{e}17$   $B_{\text{crit}} = 1\text{e}11$   $\rho_{\text{vac\_SCm}} = 7.09\text{e-}37$   $\mu_0 = 4 * \text{np.pi} * 1\text{e-}7$   $M_{\text{sun}} = 1.989\text{e}30$   $\text{kpc} = 3.086\text{e}19$**

**NGC 1316 Parameters  $M_{\text{vis}} = 3.5\text{e}11 * M_{\text{sun}}$   $M_{\text{DM}} = 1.5\text{e}11 * M_{\text{sun}}$   $M_{\text{spiral}} = 1\text{e}10 * M_{\text{sun}}$   $M_{\text{BH}} = 1\text{e}8 * M_{\text{sun}}$   $\tau = 3.156\text{e}16$  # 1 Gyr merger decay  $d_{\text{spiral}} = 50\text{e}3 * \text{kpc}$   $\rho_{\text{dust}} = 1\text{e-}21$  #  $\text{kg/m}^3$   $B = 1\text{e-}4$  #  $T$   $z = 0.005$  # NGC 1316 redshift  $k_{\text{cluster}} = 1\text{e-}12$  #  $\text{N}/M_{\text{sun}}$   $M_{\text{cluster}} = 1\text{e}6 * M_{\text{sun}}$   $\sigma_{\text{dust}} = 2\text{e}3 * \text{kpc}$**

```
def M_total(t):
    return M_vis + M_DM + M_spiral * np.exp(-t / tau)

def H_z(z):
    return H_0 np.sqrt(0.3 (1 + z)**3 + 0.7)

def F_env(t):
    F_tidal = G M_spiral / d_spiral**2
    F_cluster = k_cluster M_cluster
    return F_tidal + F_cluster

def U_i():
    rho_vac_UA = 7.09e-36
    return 1.0 (rho_vac_SCm / rho_vac_UA) 1e-8 1.0 1.01

def g_NGC1316(r, t=3.156e15): # t=100 Myr default
    M = M_total(t)
    H = H_z(z)
    F = F_env(t) / (G M / r**2)
    g_grav = (G M / r**2) (1 + H) (1 - B / B_crit) (1 + F)
    g_dust = rho_dust (4/3 np.pi r**3) G M / r**2
    dm_term = (M_vis + M_DM) (1e-5 + 3 G M / r**3)
    cosmological = Lambda c**2 / 3
    return g_grav + U_i() + cosmological + g_dust + dm_term
```

**Compute  $r_{\text{vals}} = \text{np.linspace}(1\text{e}3 * \text{kpc}, 100\text{e}3 * \text{kpc}, 100)$   $g_{\text{vals}} = [g_{\text{NGC1316}}(r) \text{ for } r \text{ in } r_{\text{vals}}]$**

```
plt.figure(figsize=(10, 6))
plt.plot(r_vals / kpc, g_vals, 'r-', linewidth=2, label='NGC 1316 MUGE Acceleration')
plt.xlabel('Radius (kpc)')
plt.ylabel('g_NGC1316 (m/s^2)')
plt.title('NGC 1316 Fornax A -- UQFF MUGE Radial Profile')
plt.legend()
plt.grid(True, alpha=0.3)
plt.savefig('ngc1316_gravity_profile.png', dpi=150)
plt.close()
print("NGC 1316 simulation complete. Profile saved to ngc1316_gravity_profile.png")
'''
```

---

## 6. Comparative Analysis

| Property       | M51 (Whirlpool)                                   | NGC 1316 (Fornax A)                       |
|----------------|---------------------------------------------------|-------------------------------------------|
| Type           | Sc spiral                                         | E/S0 elliptical                           |
| Distance       | 7.7 Mpc                                           | 75 Mpc                                    |
| Interaction    | Active (NGC 5195)                                 | Past mergers (~1-3 Gyr)                   |
| Key term       | $F_{\text{tidal}}$ (live)                         | $F_{\text{cluster}} + \rho_{\text{dust}}$ |
| Star formation | SFR $\approx 1 \text{ MM}_{\text{sun}}/\text{yr}$ | $\sim 0$ (quenched)                       |
| AGN activity   | Moderate                                          | Strong (radio lobes)                      |
| Key physics    | Density waves ( $\psi_{\text{spiral}}$ )          | Dust lane drag ( $\rho_{\text{dust}}$ )   |

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## 7. Astrophysical Significance

**M51:** The  $F_{\text{tidal}}$  term drives the prominent spiral arms visible in Hubble observations. The  $\psi_{\text{spiral}}$  density wave modulation affects star formation efficiency across the disk. NGC 5195's tidal force creates the bridge structure and enhanced SFR in the outer arms.

**NGC 1316:** The  $F_{\text{cluster}}$  term explains the deficit of low-mass globular clusters in the inner region (disrupted by tidal forces during multiple mergers). The  $\rho_{\text{dust}}$  dust lane creates a complex hydrodynamic environment near the AGN.

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## 8. Conclusion

The M51 and NGC 1316 MUGEs demonstrate the UQFF framework's ability to model both active interaction (M51) and post-merger (NGC 1316) galaxy evolution with Hubble-calibrated parameters. The included Python simulation scripts provide quantitative radial acceleration profiles ready for comparison with observational rotation curves and X-ray surface brightness profiles. These two contrasting systems validate the modularity of  $F_{\text{env}}(t)$  across diverse galactic environments.

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Session 180 continuation v5.38.

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<!-- PKG-GW-S225 -->

## Session 225 Phonon-Physics Upgrade: GW Strain Modulation

> Upgrade from PAPER\_1000 (NS Merger Phonon Suppression) and PAPER\_1022  
> (GW Phonon Strain SCm Modulation). See also PAPER\_1011-1012 for  
> GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}}\right)$$

where:

- $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$  is the phonon modulation factor
- $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$  is the SCm phonon resonance frequency
- $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left(1 + [\text{SSq}] \cdot e^{-\kappa_{i,n}/26}\right)$  is the third-order Ramanujan summation
- $\Theta$  is the Heaviside step ensuring  $H_{\text{SCm}} \geq 0.5$  (phase-transition threshold)

**Physical mechanism:** The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy  $\Delta E_{\text{BCS}}$  of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at  $M \approx 2.5 M_{\odot}$ .

**Calibration (canonical):**  $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$ ,  
 $[\text{SSq}] = 0.57$ ,  $\beta_i = 0.603$ ,  $H_{\text{SCm}} \approx 0.99$ .

<!-- PKG-CLU-S225 -->

## Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER\_1039 (SCm Galaxy Cluster Buoyancy Profile),  
> PAPER\_1041 (Cool-Core Buoyancy Balance), and PAPER\_1079 (Cooling-Flow  
> Suppression). See also PAPER\_1040 (Cluster Merger Shock), PAPER\_1044  
> (Thermal SZ Compton-y), PAPER\_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

**Hydrostatic mass bias reduction (PAPER\_1039):**

$$b_{\text{UFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20 \text{)}$$

The buoyancy pressure contributes  $P_{\text{buoy}}/P_{\text{thermal}} \approx 3 \times 10^{-4}$  at cluster cores, partially resolving the Planck SZ–CMB mass tension.

**Cool-core stabilization (PAPER\_1041/1079):** AGN feedback couples to the SCm buoyancy field via  $\dot{M}_{\text{cool}} = \dot{M}_0 (1 - \beta_i S_{26}^{(3)} \Phi)$ , suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

**Phonon frequency coupling:**  $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$  sets the temporal scale for buoyancy oscillations; the ratio  $\omega_{\text{SCm}}/\omega_{\text{sound}}$  governs the phonon transmission efficiency across the ICM.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi_{\text{NS}})^2 (\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!} \phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac, [SCm]}} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2) \phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\{\text{PAPER\_877 Axioms}\} \rightarrow \{\text{DPM + ACP}\} \rightarrow \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \rightarrow \{\text{Stage 5}\} \rightarrow U_{\text{b, seed}} \rightarrow \{\text{4 forces}\} \rightarrow F_{U_{\text{Bi}}} \rightarrow \{\text{sector E-L}\} \rightarrow \frac{\delta S}{\delta \phi_{\text{NS}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four  $U_g$  forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)



The canonical VDS ratio  $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.127 (near-threshold regime), placing it in the  $\pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 2, \quad n_{\mathrm{channel}} = 23/26$$

Since  $p_{\mathrm{DVP}} = 2$  is **sub-threshold** (threshold at  $p > 26$ ), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair  $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}} = 1)$  constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is  $10^4 \text{ yr}$  (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The  $\tanh$  saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework            | Canonical Value                                  | This Paper                          | Status                   |
|----------------------|--------------------------------------------------|-------------------------------------|--------------------------|
| VDS ratio            | $\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$ | Local sub-ratio = 0.127             | PASS                     |
| Threshold-consistent |                                                  |                                     |                          |
| DVP prime            | $p_k \in \{2,3,\dots,113\}$                      | $p_{\mathrm{DVP}} = 2$              | PASS Sub-threshold       |
| BSH layers           | 26 harmonic terms                                | $j = 1\dots 26$ , $\cos(2\pi j/26)$ | PASS Full 26D projection |
| $\kappa$ decay       | $5.0 \times 10^{-4} \text{ day}^{-1}$            | Applied in VDS exponential          | PASS Canonical           |
| $[SSq]$              | 0.57                                             | Applied in BSH saturation           | PASS Canonical           |

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                          | SM / Experiment                         | Source                              | Alignment       |
|----------------------------------|----------------------------------------------------------|-----------------------------------------|-------------------------------------|-----------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via $U_{g1}$ dipole coupling    | $1/137.036$                             | PDG 2024                            | PASS Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-5^2} \text{ m}^{-2}$ (UQFF vacuum term) | $1.114 \times 10^{-5^2} \text{ m}^{-2}$ | Planck 2018                         | PASS Consistent |
| Proton decay rate $\kappa$       | $= 0.0005/\text{day} \rightarrow \Gamma_p$ suppression   | $< 4.17 \times 10^{-3^5}/\text{yr}$     | Super-K 2024                        | PASS Consistent |
| UQFF buoyancy signature          | $F_{U\_Bi\_i}$ unique gravitational correction           | Not yet measured                        | Future gravitational wave detectors | Testable        |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{U\_Bi\_i}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{SCm}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204–225 extensions (PAPER\_1000–1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

| Paper      | Title                                                    |
|------------|----------------------------------------------------------|
| PAPER_1000 | NS Merger $F_{U\_Bi}$ Strain Suppression & BCS Gap       |
| PAPER_1001 | SMBH Binary Merger $F_{U\_Bi}$ Phonon Damping            |
| PAPER_1011 | GW170817 NS Merger $F_{U\_Bi\_i}$ 66.7% Strain Reduction |
| PAPER_1012 | GW190425 Upgraded $F_{U\_Bi\_i}$ with S26(3)             |
| PAPER_1014 | SMBH Merger Inspiral-Coalescence-Ringdown                |
| PAPER_1039 | SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model       |
| PAPER_1040 | SCm Cluster Merger Shock Mach Number Phonon Damping      |
| PAPER_1041 | SCm Cool-Core Buoyancy Balance AGN Feedback              |
| PAPER_1044 | SCm Cluster Thermal SZ Effect Compton-y Phonon           |
| PAPER_1045 | SCm Cluster Radio Relic Polarization                     |
| PAPER_1046 | SCm Cluster Lensing Mass Phonon Correction               |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression         |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum              |
| PAPER_1035 | Kilonova Buoyancy Light Curve r-Process                  |
| PAPER_1050 | MUGE $F_{U\_Bi\_i}$ Unified 9-System Synthesis           |
| PAPER_1075 | 3D Volumetric MUGE Gravitational Field Generator         |

16 cross-reference(s) identified.

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## Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.  
 > Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations,  
 > see PAPER\_840/851/852/855.

### S204.1 Kozima-UQFF LENR Integration

| Module                      | Purpose                                    | Key Result                                 |
|-----------------------------|--------------------------------------------|--------------------------------------------|
| fneutron_s26_coupling.py    | F_neutron x S_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS       |
| kozima_scm_cross_section.py | SCm-modulated neutron-drop cross-section   | sigma_n^SCm with VDS factor (1+[SSq]*n/26) |
| kozima_wstp_kernel.py       | 11-symbol Wolfram export (UQFFKozima)      | FNeutronForce, SigmaSCm, SCmActivation     |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$   
 where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] * n/26)$

### S204.2 Ramanujan 26-State Summation

| Module                   | Purpose                                       | Key Result                |
|--------------------------|-----------------------------------------------|---------------------------|
| ramanujan_polylog_s26.py | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| s26_wstp_kernel.py       | 8-symbol Wolfram export (UQFFS26)             | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * Li_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module            | Purpose                                | Key Result                  |
|-------------------|----------------------------------------|-----------------------------|
| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:**

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$
- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                       | Purpose                                   | Key Result                                 |
|------------------------------|-------------------------------------------|--------------------------------------------|
| ramanujan_pi_uqff.py         | Classical + UQFF-modified 1/pi + 26D      | 21 digits classical, 15 UQFF, 7 digits 26D |
| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2})/9801 \sum R_n (1103+26390n) W_{26}(n) / C_{26}$   
 where  $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{\text{pai}} * n/26)]$

### S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_ScM     | 0.99                                  | SCm manifold completeness     |
| rho_ScM   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_ScM | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation: all modules operational in* CondensedPhysics.py, CondensedPhysics2.py, MAIN\_{1\\_CoAnQi}.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).

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**G/c DERIVATION NOTE (appended 2026-07-22, UNIFIED REGISTRY R2 corpus pass)**

This paper uses  $G = 6.674\text{e-}11$  (CODATA form) and  $c = 3\text{e}8$ -family literal as published. Per the Unified Registry (R1-adjudicated canonical routes, 2026-07-22):

- **G (gravitational constant)**: canonical route **PAPER\_593** — parameter-free

$G_{\text{UQFF}} = (2\pi \cdot 26^3 \cdot \Phi_{\text{res}} / (SSq^3 \cdot (26!)^2)) \cdot v_F \blacksquare / (E_0 \cdot f_{\text{THz}}) = 6.66899 \times 10 \blacksquare^{11}$  (0.08% vs observed).

- **c (speed of light)**: canonical route **PAPER\_592** — parameter-free

$c_{\text{UQFF}} = (26 \cdot 4\pi / \Phi_{\text{res}}) \cdot v_F = 2.995 \times 10 \blacksquare \text{ m/s}$  (0.13% vs observed;  $v_F$  Fermi anchor, c-independent).

Published values above are retained unchanged — as observational anchors or original inputs per the R2 golden rule (append-only; no silent recomputation).

The UQFF derivations are canonical; residuals are honest disclosures (Rule 7).

Registry: UNIFIED\_REGISTRY.csv | Program: UNIFIED\_REGISTRY\_PROGRAM\_PLAN.md

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